

3/2 Group Meeting

11

Generalized Kummer varieties
and Beaville - Mukai systems.

Motivation behind this talk

Consider a (cpt) Hyper Kähler (one of the irreducible
factors arising in BB
decomp for CY)

admitting a Lagrangian fibration (generalization of
an elliptic fibration on K3)

$$\pi: X \rightarrow B. \iff \tilde{\pi}: \tilde{X} \rightarrow B \quad \tilde{\pi} \approx \text{Abel var fib.}$$

We want geom categorical relationship between

the total space and a suitable dual fib = generically fiberwise

Motivated from SYZ (Strominger-Yau-Zaslow) conj dual
for (strict) CY. mirror symm.

and ii) Bridgeland-Macriocia (99, 02)

for ellip fib on surf. relative Jacobian.

$$\pi: X \rightarrow C, \quad \hat{\pi}: \hat{X} := \text{Jac}(X/C) \rightarrow C$$

$$\Rightarrow D^b(X) \simeq D^b(\hat{X})$$

Some attempts have been successful

• One way of construction of dual ^{nice} in "Alg geom seminar"

→ Yoon-Joo Kim (This Fri. talk)

• K3^[2] case by Justin Sawon (3/27 talk)

Recent project

We work on Kum^[2]

Today we overview Kum^[m]-type IHS

§ Preliminaries

[2]

Def cpt Hyper Kähler (a.k.a. irreducible holomorphic symplectic mfd)

- X : cpt Kähler mf s.t.
- i) $H^0(X, \Omega_X^2) = \mathbb{C} \sigma$ σ : everywhere non-vanishing global 2-form.
 - ii) X : simply conn.

generalization of K3.

(exclude Ab)

Recall int. form on K3.

play important role.

Thm-Def

X : HK

- i) $\exists$: primitive ($\nexists m > 1$ s.t. $m \sigma$ is indivisible):
nondeg symm bilinear form

$$\vartheta: H^1(X, \mathbb{Z}) \otimes H^2(X, \mathbb{Z}) \rightarrow \mathbb{Z} \quad (\text{Beauville-Bogomolov-Fujiki form.})$$

- ii) $\exists$: const. $C_X \in \mathbb{Q}_{>0}$ (Fujiki const)

$$\int_X \alpha^{2n} = C_X \frac{(2n)!}{2^n n!} \vartheta(\alpha, \alpha)^n \quad \alpha \in H^2(X, \mathbb{Z})$$

Fujiki relation.

this comes from polarized form

$$\int_X x_1 \cdots x_{2n} = C_X \sum_{\sigma \in S_{2n}} \vartheta(x_{\sigma(1)}, x_{\sigma(2)}) \cdots \vartheta(x_{\sigma(2n-1)}, x_{\sigma(2n)})$$

Thm (Matsushita)

Def X : HK of dim $2n$.

Lag fib of X to a sm base is a hol surj map

$\pi: X \rightarrow B$ w/ conn. fiber a cpx prj mfd B .

$$0 < \dim B < \dim X.$$

Def X : HK of dim $2n$ (not necessarily cpt) $\rightarrow$ Hitchin system.

Lag fib: proper map $h: X \rightarrow B$ onto a mfd B .

s.t. gen fiber F of h is Lagrangian. (F : conn, of dim n , $\omega|_F = 0$)

Surprising fact (Hwang '08, Guck-Lehn '14)

X : HK of dim $2n$ $\pi: X \rightarrow B$ Lag fib w/ B : sm

$\Rightarrow B \cong \mathbb{P}^n$ (conjectured B : normal $\leadsto B \cong \mathbb{P}^n$)

Note $H := \pi^* \mathcal{O}_B(1)$ pull back of an ample

[3]

$$h = c_1(H) \in NS(X)$$

$$\Rightarrow \int_X \underbrace{h^{2n}}_n = \text{const. } \vartheta(h)^n \Rightarrow \vartheta(h) = 0$$

$$0 \leq B := \dim n$$

$\exists h$: isotropic class

elliptic
for $K3$)

h^h : fiber class.

(Recall $F^2 \sim F \cdot F' \sim 0$

F : fiber class

Conj (Generalized Abundance conj for HK, HK-STZ)

X : HK, $\exists L$: nef & isotrivial $\Rightarrow L$: semiample

(not big) $\Rightarrow |mL|$ def a Lag fib for $m \gg 0$

X : HK, $\exists L$: isotrivial $\Rightarrow X$ admits rational Lag fib.

ie. $\exists X' : \text{IHS} \xrightarrow{\text{w/ Lag fib}} \phi: X' \xrightarrow{\sim} X$ bir equiv
for ϕ : fib over B .

This true for $K3^{[n]}$: Sawon '07, Markshovich '08.

$Kum^{[n]}$: Gulbrandsen '07, Yoshioka.

↓ simpler

Using (twisted) FM & BM system (later)

• using twisted Univ. sheaf by Brauer

(Mayer-Macri (21)
wall-crossing
Thm 1.5)

Known example 4 known def type.

1) $K3^{[n]}$ -type

• Hilb scheme of points of $K3$. $\dim \mathbb{C} H^0(\mathcal{O}_Z)$

$$S^{[n]} := \text{Hilb}^n(S) = \{Z \subset S \mid \dim Z = 0, \text{length}(Z) = n\}$$

• $M_H(v) \begin{cases} v: \text{primitive} \\ H: v\text{-generic} \end{cases} \Rightarrow M_H(v) : K3^{[n]} \text{-type } n = \frac{\langle v, v \rangle + 2}{2}$

3). OG-10

Key $\text{Hilb}^n \cong M_S(1, 0, 1-n)$ using Ideal sheaf $\tilde{\mathcal{I}}_Z$

v : non-primitive $\Rightarrow$ sm symp resol $\Rightarrow$ OG-10

$$(\text{td}_X = (1+w))$$

$$ch(\tilde{\mathcal{I}}_Z) = (1, 0, -n)$$

2) Kumⁿ - type.

- A : ab surf.

- A^{n+1} has torus comp (i.e. not simply conn.)

$t_a: A \rightarrow A$ parallel translation
 $t \mapsto t+a$

$T_a: A^{n+1} \rightarrow A^{n+1} \ni$ freeness
 $\{t\} \mapsto \{t+a\}$

- $A^{(n+1)} := A^{n+1} / G_{n+1}$: symm. prod
 (space of cycle)

- Hilbert-Chow map gives resolution

$$\left[\begin{array}{l} P: A^{(n+1)} \rightarrow A^{(n+1)} \\ Z \mapsto \sum_{x \in \text{supp}(Z)} \text{length}(\mathcal{O}_{Z,x}) \cdot x \end{array} \right]$$

target sch. str.
 only record multi & supp.

- $\Sigma: A^{(n+1)} \rightarrow A$ summation
 (G_{n+1} -inv $\Rightarrow$ well defined)

- Fact $(\Sigma \circ P): A^{(n+1)} \rightarrow A$ albrese map
 isotrivial fib & $\text{Kum}^n(A) := (\Sigma \circ P)^{-1}(0)$

classical Kummer

$A^2: x+y=0 \in A/\{\pm 1\}$ "zero-sum antigravitation of points"

w/ 16 2-torsion
 resul.

• $M_H(v): v$: primitive $H: v$ -generic

$\Rightarrow$ holomorphic sym var (Not nec. simply conn)

How to fill out torus comp?

[Yoshioka]

$\text{Alb} = M_H(v) \rightarrow A \times \hat{A}$ $\hat{A} = P_{\text{ic}}^0(A)$

$E \mapsto (\det \bar{\mathbb{P}}_P(E) \otimes \det \bar{\mathbb{P}}_P(E_0)^V, \det E \otimes \det E_0^V)$

$E_0: t \cdot x$, P : Poincaré sheaf on $A \times \hat{A}$

$\bar{\mathbb{P}}_P$: FM w.r.t P .

$\Rightarrow \text{Alb}^{-1}(0,0)$: gen Kum type.

4) $\text{Alb}: M \xrightarrow{\text{resul}} \bar{M} \rightarrow A \times \hat{A}$ for v : n -prime $\Rightarrow$ OGB.

Some topological invariant

Betti (evidence for the fact det family different.)

- $b_2(X)$

$$= \begin{cases} 23 & K3^{17} \\ 24 & OG10 \\ 7 & Kum^{17} \\ 8 & OG6 \end{cases}$$

Antonio
Beauville - Rapagnetta

• BB lattice
($H^2(X, \mathbb{Z}), g$) use $\begin{cases} H^2(M_H(u), \mathbb{Z}) \simeq U^\perp & (Yoshioka) \\ H^2(K_H(u), \mathbb{Z}) \simeq U^\perp & Kum^n \end{cases}$

$$\cong \begin{cases} U^{\oplus 3} \oplus E_8(-1)^{\oplus 2} \oplus \langle -2(n-1) \rangle & K3^{17} \\ U^{\oplus 3} \oplus E_8(-1)^{\oplus 2} \oplus A_2(-1) & OG10 \\ U^{\oplus 3} \oplus \langle -2(n+1) \rangle & \uparrow \text{exc. div. } Kum^{17} \\ U^{\oplus 3} \oplus \langle -2 \rangle^{\oplus 2} & OG6 \end{cases}$$

• Fujiki const.

$$= \begin{cases} 1 & K3^{17}, OG10 \\ n+1 & Kum^{17} \\ 4 & OG6 \end{cases}$$

(*) $U = (1, 0, 1-n)$
 $(v, c, s) \in U^\perp$
 $\Leftrightarrow -v(1-n) - s = 0 \Rightarrow s = (n-1)v$
 $H^2(S, \mathbb{Z}) \oplus \mathbb{Z} \simeq U^\perp$
 $(c, v) \mapsto (v, c, (n-1)v)$
 $w/ \delta^2 = (1, 0, n-1)^2 = -2(n-1)$
 $\uparrow$ new direction (v direction)

- Topological Euler char.

• Göttsche (90) $K3^{17}$

$$\sum_{n \geq 0} e(S^{17}) g^n = \prod_{m \geq 1} (1 - g^m)^{-24}$$

$$e(K3^5) = 176256$$

$$e(Kum^n(A1)) = (n+1)^3 \sigma(n+1)$$

sum of divisors.

$$e(Kum^3) = 256$$

Antonio

• Rapagnetta '07 $e(OG6) = 1920$

Mozgovyy '06 $e(OG10) = 176904$

sm fiber of Lag fib $\xrightarrow{\text{Matsushita}}$ abelian var.

what's the polarization type

If fibers are NOT ppav

$\Rightarrow$ sm fib of $\overline{\text{Pic}}^0(X/\beta)$ & corresp sm fib

only isogenous

{torsion free rank 1 sheaves
of deg 0 on fibers of X }

Polarization

(Wieneck '16, '18 KS, Kum)
(Mongardi - Rapagnetta '21 O G 6.)
(Mongardi - Onorati '22 O G 10)

$(1, \dots, 1) \subset \text{O G 10, KS}^{[n]}$

$(1, \dots, 1, d, \frac{n+1}{d}) \subset \text{Kum}^{[n]}$

$(1, 2, 2) \subset \text{O G 6}$

$d = \text{div}(h)$ divisibility
 $= \gcd \{ \delta(x, h) \mid x \in H^2(X, \mathbb{Z}) \}$
 $h := c_1(\pi^* \mathcal{O}_{\mathbb{P}^1}(1))$

use complementary ab. var.
see Birkenhake-Lange

Ideg

$$K(C) = \ker(J(C) \rightarrow S)^0$$

$$(K(C), \hat{S}) \subset (J(C), \mathbb{Q})$$

ppav

$\simeq Z: C \hookrightarrow S$ & univ map of
Abel-Jacobi map

$$\begin{array}{ccc} x & \hookrightarrow & S \\ \downarrow \alpha_C & & \downarrow t = \tau(a) \\ \mathcal{O}(x-C) & \xrightarrow{J(C)} & S \end{array}$$

Beaville Mukai system

[17]

Nice testing ground for Lag fib.

S : surf (K3 or Abelian)

$C \subset S$: sm curve

$|C|$: complete lin sys. of dim $h^0 - 1$.

$\mathcal{C} = \{D \in |C|\}$ family of curves.

then $\boxed{\overline{\text{Pic}}^0(\mathcal{C}/|C|) \longrightarrow |C|}$ BM-system.

$\overline{\text{Pic}}^0 D \xrightarrow{\quad} \overset{w}{D}$
unpctified Pic sch.

Idea of Sawon/Markushevich

for gen. ab. var for K3^[n]. $g(L) = 0 \Rightarrow \exists$ rat Lag fib.

Recall (Toshioka) $S^{[n]} = M_H(v_0)$ $v_0 = (1, 0, 1-n)$

$$\underline{p(S) = 1} \quad H^2(S^{[n]}) \cong v_0^\perp$$

$$L \longleftrightarrow w$$

$$g(L) = 0 \Leftrightarrow w^2 = 0$$

$$\Leftrightarrow \begin{cases} \langle w, v_0 \rangle = 0 \\ \langle w, w \rangle = 0 \end{cases}$$

$$\langle (r, c, s), (1, 0, 1-n) \rangle = -r(1-n) - s = 0$$

$$\langle (r, c, s), (r, c, s) \rangle = c^2 - 2rs$$

$$\Rightarrow c^2 = 2r^2(n-1)$$

in particular $r=1$.

$$\therefore g(L) = 0 \Leftrightarrow \exists C \text{ curve genus } n.$$

set

$$w = (1, c, n-1)$$

$$S' = M_S(w) : \text{K3 again}$$

$$\bar{\mathbb{I}} : D^b(S) \xrightarrow{\sim} D^b(S', \alpha) \quad \text{twisted FM}$$

$$w \mapsto (0, 0, 1)$$

gives equiv

$$\boxed{v_0 \mapsto (0, D, k)}$$

$M_{S'}(0, D, d) \rightarrow |D| : \text{BM system}$

take supp
 $d \quad \text{Hilb}^n(S) \dashrightarrow M_{S'}(0, D, d)$
 check WJT

Note

$\overline{\text{Pic}}^0(D/|D|):$

= torsion-free rk 1
 of deg zero on fiber.

Same idea for abelian & gen Kumm. (kill out torsion comp)

$\Rightarrow$ Gulbrandsen, Yoshioka

or Another / more refined proof
 via wall-crossing

pure 1-dim

recent prog

$$F \cong \mathbb{Z}_G L$$

$$\Rightarrow \chi(F) = \deg(L) + 1 - g(C).$$

We want work on $\text{Kum}^2 \quad |L| \approx \mathbb{P}^2$

so we need work on (A, L) (1,3)-pol. ab surf.

discrim. locus in $|L|$

$\leadsto$ catanese-serres